# SaddlepointExpansions for MATLAB

User manual

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Version 0.2.3, November 2026

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# 1 Introduction

SaddlepointExpansions for MATLAB computes saddlepoint expansions of any order for the distribution of a random vector from its cumulant generating function alone. It implements the expansions of Broda and Kan, "Uniform saddlepoint expansions for multivariate tail probabilities":

- the orthant probability $\mathrm{P}(X \geq x)$ of a random vector $X$ of any dimension $d$, with lower and mixed corners, and with the inclusion–exclusion step for saddlepoints that have negative components;
- the density of $X$ in any dimension;
- the joint density and the joint distribution function of a vector of ratios $R = (X_1/Y_1, \ldots, X_d/Y_d)$ with positive denominators, including ratios with a common denominator;
- in one dimension, the expansions of Daniels (density), Hauschildt and Robinson (distribution function) and Rice (density of a ratio).

It also implements the expansions of Bao, Broda and Kan, "Inversion formulas and saddlepoint expansions for Studentized means", for the density, the distribution function and the quantiles of a Studentized mean $Q = X/\sqrt{Y}$ with $Y > 0$, from the joint cumulant generating function of $(X, Y)$ (§ 8). The approximation of order $M$ has relative error $O(n^{-M})$, where $n$ is the large parameter (a sample size, a number of degrees of freedom).

The user writes the cumulant generating function $K$ as an ordinary MATLAB function. Its derivatives of all orders are generated by Taylor-mode automatic differentiation (the class `tps`, truncated multivariate power series with overloaded arithmetic and elementary functions), directly in the whitened coordinates in which the expansions are evaluated, so that no derivative has to be written by hand and none is computed by finite differences. The expansions themselves are computed by the MATLAB programs that accompany the paper (folder `engine`), which can also be called directly with closed-form derivatives.

The package is the MATLAB counterpart of the Julia package `SaddlepointExpansions.jl` by Simon Broda (`https://github.com/s-broda/SaddlepointExpansions.jl`), which also has a Python version (`https://github.com/s-broda/saddlepoint-expansions`). The packages have the same interface and share a set of reference values, on which they agree to about $10^{-13}$ relative (§ 13). Where double precision runs out, the Julia package recomputes in variable precision and this package in double-double arithmetic, about 32 digits (§ 9.1).

Theorem and corollary numbers refer to the version of the paper of November 2026: Theorem 1 (orthant probability), Theorem 2 (inclusion–exclusion), Theorem 3 (density), Theorem 4 and Corollary 3 (densities of ratios), Corollaries 1, 2, 4 and 5 (Daniels, Hauschildt–Robinson, the distribution function of a ratio, Rice).

**Requirements.** MATLAB (the package was tested with R2025b) and the Statistics and Machine Learning Toolbox, whose `normcdf` and `normpdf` are used by the univariate engine. Nothing else is needed.

**Installation.** Copy the folder anywhere and run `spx_setup`, which adds the folder and its `engine` subfolder to the path. `help SaddlepointExpansions` lists the functions, `examples/demo_spx.m` runs the examples of this manual in a few seconds and `tests/test_spx.m` the tests of § 13 in about a minute.

## 2 A first example

Let $X = (U_0 + U_1, U_0 + U_2)$ with $U_0, U_1, U_2$ independent unit exponential variables, the example of Kolassa and Li (2010), and let $\bar{X}$ be the mean of $n = 5$ independent copies of $X$. Then

```
spx_setup
K = CGF(@(t) -log(1 - t(1)) - log(1 - t(2)) - log(1 - t(1) - t(2)), 2, ...
        'domain', @(t) t(1) < 1 && t(2) < 1 && t(1) + t(2) < 1);
Kn = sample_mean(K, 5);               % the cgf 5 K(t/5) of the mean
r = tailprob(Kn, [3 3], 'order', 4);  % P(mean1 >= 3, mean2 >= 3)
r.F'                                  % 0.02142343 0.02214365 0.02215413 0.02215193
r.tau                                 % the saddlepoint
s = density(Kn, [3 3], 'order', 4);   % the joint density of the mean at (3, 3)
```

The exact probability is 0.02215196; the approximations of orders one to four have relative errors $-3.3 \times 10^{-2}$, $-3.8 \times 10^{-4}$, $9.8 \times 10^{-5}$ and about $-1.4 \times 10^{-6}$. The same cumulant generating function is in the library as `exponential_sums_cgf([1 0; 1 1; 0 1], 'n', 5)`.

## 3 Conventions

**The large parameter.** The large parameter $n$ is absorbed in the cumulant generating function: the expansions are those of the vector whose cumulant generating function is supplied. For the mean of $n$ copies one supplies $nK(t/n)$, which `sample_mean(K, n)` constructs; the density returned is then that of the mean.

**Orders.** Every expansion function takes `'order', M` (default three) and returns the approximations of orders $1, \ldots, M$ in a column vector. The approximation of order $M$ has relative error $O(n^{-M})$. For a density it keeps the terms $m = 0, \ldots, M-1$ of the series in powers of $n^{-1}$, and for an orthant probability the terms $m = 0, \ldots, 2M-1$ of the series in powers of $n^{-1/2}$. For densities, order one is Daniels's first-order approximation; for orthant probabilities, the leading term $m = 0$ is the approximation of Kolassa (2003).

**Results.** The expansion functions return a structure with the approximations (`F` or `f`), the saddlepoint (`tau`, or `shat` for ratios), a `status` string (`'ok'` or a warning about the numerical evaluation) and `info`, the diagnostics of the underlying engine program. The diagnostics measure the numerical quality of the evaluation, such as cancellation among the terms or the change with half the quadrature nodes; they do not measure the asymptotic error of the expansion, which is what the order controls.

**Vectors.** Points may be row or column vectors. The argument `t` of a cumulant generating function is always a $d \times 1$ column vector.

## 4 Cumulant generating functions

### 4.1 The class CGF

```
K = CGF(fun, d)
K = CGF(fun, d, 'domain', pred)
K = CGF(fun, d, 'Taylor', tfun)
```

`fun` is a function of a $d \times 1$ vector `t` returning $K(t)$, also for $d = 1$. `pred` is an optional predicate `@(t) logical` describing the domain, and `tfun` an optional function that returns Taylor polynomials in closed form (§ 4.5).

The function is evaluated on vectors of truncated power series to generate its derivatives, so it must be written with operations that the class `tps` supports:

- `+`, `-`, `.*`, `./`, `.^`, `*` and `/` by numbers, products `A*t` and `t.'*A` with constant matrices, `sum`, `prod`, `dot`, indexing `t(j)`, `t(1:k)`, concatenation, `reshape`, `diag`;
- `exp`, `expm1`, `log`, `log1p`, `sqrt`, `sin`, `cos`, `tan`, `sinh`, `cosh`, `tanh`, `atan`, `abs`, and `sinhc`, `sincu` of the package;
- comparisons (`<`, `<=`, ...), which compare the constant terms, so that code with tests on the argument can be differentiated.

Functions of `t` that call compiled numerical routines (`det`, `inv`, `chol`, `eig`, `expm`, ...) cannot be differentiated this way; § 4.4 shows how to write determinants and inverses.

**The domain.** Outside the domain, `fun` may raise an error with identifier `'tps:domain'` (the logarithm and the square root of a nonpositive number do so automatically) or return a nonfinite value; both are treated as "outside the domain", and the saddlepoint iteration steps back. When the formula is finite outside the true domain, as $1/(1-t)$ is for $t > 1$, the predicate `'domain'` is needed. It is also cheaper than an error.

## 4.2 The library

| constructor | distribution |
|---|---|
| `normal_cgf(mu, Sigma)` | $N(\mu, \Sigma)$ (the expansions are exact at order one) |
| `gamma_cgf(a, b)` | gamma with shape $a$ and scale $b$ (default one); `gamma_cgf(1, b)` is the exponential distribution with mean $b$ |
| `ncx2_cgf(m, lam)` | noncentral chi-squared $\chi^2_m(\lambda)$ |
| `exponential_sums_cgf(A, 'n', n, 'shift', l)` | the mean of $n$ independent copies of $A^\mathsf{T}U + l$, $U$ a vector of independent unit exponential variables |
| `quadratic_forms_cgf(Mats, 'Sigma', S, 'mu', mu, 'scale', c)` | the quadratic forms $z^\mathsf{T}A_i z$, $i = 1, \dots, d$, in $z \sim N(\mu, \Sigma)$, `Mats = {A_1,...,A_d}`, summed over $c$ independent copies |
| `wishart_diag_cgf(V, n)` | the diagonal of a Wishart matrix $W_d(n, V)$ |
| `common_component_cgf(m, lam)` | $Q_j = X_j + X_0$, $j = 1, \dots, d$, with independent $X_i \sim \chi^2_{m_i}(\lambda_i)$, `m = [m_1 ... m_d m_0]` |
| `uniform_cgf(a, b)` | uniform on $[a, b]$ (default $[-1, 1]$) |

The cumulant generating function of quadratic forms is

$$K(t) = -\frac{1}{2}\log\det(I - 2C) + \nu^\mathsf{T} C (I - 2C)^{-1}\nu, \qquad C = \sum_{i=1}^{d} t_i \Sigma^{1/2} A_i \Sigma^{1/2}, \qquad \nu = \Sigma^{-1/2}\mu,$$

on the set where $I - 2C$ is positive definite. Its Taylor polynomials are computed in closed form by the trace recursion of the paper, which is much faster than automatic differentiation for matrices of large order.

## 4.3 Transformations

| function | cumulant generating function of |
|---|---|
| `reflect(K, s)` | $(s_1X_1, \dots, s_dX_d)$, $s_j = \pm 1$; `reflect(K)` is that of $-X$ |
| `marginal(K, idx)` | the components `idx` of $X$ |
| `linear_combination(K, B)` | $B^\mathsf{T}X$ for a $d \times k$ matrix $B$, that is $K(Bu)$ |
| `sample_mean(K, n)` | the mean of $n$ independent copies, $nK(t/n)$ |
| `shift(K, c)` | $X + c$ |
| `K1 + K2` | the sum of two independent vectors of the same dimension |

The transformations keep closed-form Taylor polynomials: a linear combination of quadratic forms, for example, is still differentiated by the trace recursion.

### 4.4 Writing your own

Removable singularities are the one trap: $\sinh(t)/t$ is $0/0$ at $t = 0$, where the saddlepoint iteration starts. Use `sinhc` and `sincu`, which switch to the power series there; the library defines the uniform distribution as `log(sinhc(t))`. Two patterns:

```
% independent parts, one per component, plus a common part
K = CGF(@(t) -sum(m(:)/2.*log(1 - 2*t)) - m0/2*log(1 - 2*sum(t)), d, ...
        'domain', @(t) all(t < 1/2) && sum(t) < 1/2);

% a log-determinant: a Cholesky factorization written in plain arithmetic
function v = logdet_cgf(t, B1, B2)
n = size(B1, 1); Q = eye(n) - 2*(t(1)*B1 + t(2)*B2);
L = Q(1, 1)*zeros(n);                % a series array of the right class
for j = 1:n
   s = Q(j, j); for k = 1:j-1, s = s - L(j, k)^2; end
   L(j, j) = sqrt(s);
   for i = j+1:n
      w = Q(i, j); for k = 1:j-1, w = w - L(i, k)*L(j, k); end
      L(i, j) = w/L(j, j);
   end
end
v = -sum(log(diag(L)));
end
```

The test file `tests/test_spx.m` differentiates the complete cumulant generating function of a pair of quadratic forms written this way and checks it against the trace recursion. For quadratic forms in normal variables, `quadratic_forms_cgf` is faster.

### 4.5 Closed-form Taylor polynomials

The option `'Taylor', tfun` supplies a function `tfun(t, N, M)` that returns the Taylor polynomial of degree $N$ in $v$ of $K(t+Mv)$ as a scalar `tps` in `size(M, 2)` variables (`M = []` stands for the identity). It is then used instead of automatic differentiation; `fun` is still used for the values of $K$. The polynomial is built from its coefficients with `tps(C, [1 1], k, N)`, where `C` holds the coefficients of the monomials in the order of `tps.tables(k, N).mi` (§ 9). `quadratic_forms_cgf` is an example.

## 5 Orthant probabilities

```
r = tailprob(K, x, 'order', M)                  % P(X >= x)
r = tailprob(K, x, 'order', M, 'lower', true)   % P(X <= x)
```

```
r = tailprob(K, x, 'order', M, 'signs', s)      % e.g. s = [1 -1]: P(X1 >= x1, X2 <=
    x2)
```

`r.F` holds the approximations of orders $1, \dots, M$ and `r.tau` the saddlepoint, $K'(\hat{\tau}) = x$. Lower and mixed corners are computed as upper orthant probabilities of the reflected vector.

**Saddlepoints with negative components.** The expansion of Theorem 1 applies to saddlepoints in the closed positive orthant $\hat{\tau} \geq 0$. When some components are negative, `tailprob` writes the probability by the inclusion–exclusion identity of Theorem 2 as a signed sum of orthant probabilities of marginals with some components negated, each with a saddlepoint in its closed positive orthant, recursively, and expands every term (§ 10). This keeps the relative error $O(n^{-M})$; a direct evaluation of the formula at such a saddlepoint would have the wrong exponential order. The saddlepoint reported is that of the original problem.

**Dimensions.** In one dimension the expansion is that of Hauschildt and Robinson (Corollary 2), computed directly for the upper tail. In two dimensions the integrals of the expansion are computed by Gauss–Legendre quadrature (`spa_orthant2`, `spa_bi`), in three by Gauss quadrature on a conditioned grid (`spa_tri`), and in four or more by randomized quasi-Monte Carlo after Genz's sequential conditioning (`spa_mv`); `r.info.err` is then an error estimate (in four or more dimensions a standard error). A trivariate probability takes about a second at order three and tens of seconds at order four or five, mostly in the inclusion–exclusion terms.

**Options.** Name-value pairs other than `'order'`, `'lower'` and `'signs'` are passed to the engine: `'nG'` (Gauss–Legendre nodes per coordinate), `'qtol'` (quadrature tolerance), `'nGmax'`, `'method'` (`'quad'` or `'qmc'`), `'nqmc'`, `'nshift'`, `'seed'` and, in two dimensions, `'logscale'`, `c`, which returns the probability scaled by $e^{-c}$ for probabilities below the range of double precision. See the help of `spa_bi`, `spa_tri` and `spa_mv`.

# 6 Densities

```
r = density(K, x, 'order', M)
```

`r.f` holds the approximations of orders $1, \dots, M$ of the density of $X$ at $x$ (Theorem 3; Daniels's expansion, Corollary 1, in one dimension). The expansion is a finite sum of Hermite polynomials at zero and needs no integration in any dimension. In three or more dimensions the terms of high orders can cancel by many orders of magnitude; `r.info.kappa_pdf` reports the cancellation (§ 11).

# 7 Ratios

## 7.1 Usage

Let $R = (X_1/Y_1, \dots, X_d/Y_d)$ with $Y_j > 0$. The cumulant generating function `K` is that of the vector $(X_1, \dots, X_d, Y_1, \dots, Y_d)$, of dimension $2d$, numerators first. For ratios with a common denominator $Y$ it may instead be that of $(X_1, \dots, X_d, Y)$, of dimension $d + 1$. The number of ratios is the length of the point `r0`.

```
r = ratio_density(K, r0, 'order', M)                % density of R at r0
r = ratio_tailprob(K, r0, 'order', M)               % P(R >= r0)
r = ratio_tailprob(K, r0, 'order', M, 'lower', true)  % P(R <= r0)
r = ratio_tailprob(K, r0, 'order', M, 'signs', s)   % a mixed corner
```

`r.f` or `r.F` holds the approximations of orders $1, \dots, M$, and `r.shat` the saddlepoint of $K_W$ (§ 10.4). For one ratio the density is Rice's expansion (Corollary 5), and for several the expansion of Theorem 4 and Corollary 3, whose term of order one is the approximation of Butler and Paolella (1998); the distribution function is the orthant probability of § 5 for $W = X - r_0 \circ Y$ (Corollary 4 for one ratio), with all its options.

## 7.2 Examples

**Sample autocorrelations.** With $z \sim N(0, I_N)$, $M = I_N - 1_N 1_N^\mathsf{T}/N$ and $(A_j)_{ts} = 1/2$ for $|t - s| = j$, the sample autocorrelations at lags one and two are $R_j = z^\mathsf{T} M A_j M z / z^\mathsf{T} M z$, ratios with the common denominator $z^\mathsf{T} M z$. For $N = 50$ at $r_0 = (-1, -1)/\sqrt{50}$:

```
N = 50; M = eye(N) - ones(N)/N;
A = @(j) (diag(ones(N-j, 1), j) + diag(ones(N-j, 1), -j))/2;
K = quadratic_forms_cgf({M*A(1)*M, M*A(2)*M, M});      % dimension d + 1 = 3
r0 = [-1 -1]/sqrt(N);
f = ratio_density(K, r0, 'order', 5).f
F = ratio_tailprob(K, r0, 'order', 5, 'lower', true).F
```

The relative errors of orders one to five, against the exact values of the paper, are

| order | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| density | $-1.4 \times 10^{-3}$ | $7.3 \times 10^{-5}$ | $-1.0 \times 10^{-5}$ | $2.2 \times 10^{-6}$ | $-6.4 \times 10^{-7}$ |
| distribution function | $-3.7 \times 10^{-2}$ | $1.5 \times 10^{-4}$ | $1.4 \times 10^{-5}$ | $5.8 \times 10^{-7}$ | $-1.6 \times 10^{-7}$ |

These are the values of the corresponding row of the table of sample autocorrelations in the paper, which were computed by a dedicated program.

**Gamma ratios with a common denominator.** If $X_1, \dots, X_d, Y$ are independent with $X_j \sim \text{Gamma}(a_j)$ and $Y \sim \text{Gamma}(b)$, the ratios $R_j = X_j / Y$ have the inverted Dirichlet density

$$f_R(r) = \frac{\Gamma(a_1 + \cdots + a_d + b)}{\Gamma(b) \prod_{j=1}^d \Gamma(a_j)} \frac{\prod_{j=1}^d r_j^{a_j - 1}}{(1 + r_1 + \cdots + r_d)^{a_1 + \cdots + a_d + b}}.$$

```
a = [3 4 5]; b = 6;
K = CGF(@(t) -sum(a(:).*log(1 - t(1:3))) - b*log(1 - t(4)), 4, 'domain', @(t) all(t <
    1));
f = ratio_density(K, [0.6 0.5 0.9], 'order', 5).f
```

The relative errors of orders one to five are $7.7 \times 10^{-2}$, $-3.1 \times 10^{-3}$, $-1.0 \times 10^{-4}$, $1.7 \times 10^{-5}$ and $3.6 \times 10^{-6}$. Multiplying the shapes by $n$ shows the rate $O(n^{-M})$; for $d = 2$ at $r_0 = (0.6, 0.5)$:

| $n$ | order 1 | order 2 | order 3 | order 4 |
|---|---|---|---|---|
| 1 | $5.8 \times 10^{-2}$ | $-1.8 \times 10^{-3}$ | $-1.2 \times 10^{-4}$ | $1.3 \times 10^{-5}$ |
| 2 | $2.8 \times 10^{-2}$ | $-4.2 \times 10^{-4}$ | $-1.6 \times 10^{-5}$ | $6.6 \times 10^{-7}$ |
| 4 | $1.4 \times 10^{-2}$ | $-1.0 \times 10^{-4}$ | $-2.0 \times 10^{-6}$ | $3.7 \times 10^{-8}$ |
| 8 | $7.0 \times 10^{-3}$ | $-2.5 \times 10^{-5}$ | $-2.5 \times 10^{-7}$ | $2.2 \times 10^{-9}$ |

Each doubling of $n$ divides the error of order $M$ by about $2^M$.

### 7.3 Cost

The density of several ratios needs one Taylor polynomial of $\mathcal{K}(s, -r \circ s + \tau)$ in the $2d$ variables $(u, \tau)$ of § 10.4, of degree $\max(2M, 2M - 2 + d)$. For two or three ratios every order up to five takes one or two seconds, also for quadratic forms in $N = 50$ variables; for four ratios order four takes several seconds. The distribution function costs what an orthant probability of dimension $d$ costs (§ 5).

## 8 Studentized means

Let $Q = X/\sqrt{Y}$ with $Y > 0$, and let `K` be the joint cumulant generating function of $(X, Y)$, of dimension two (numerator first). The expansions of Bao, Broda and Kan are

```
r = studentized_density(K, q, 'order', M)                  % density of Q at q
r = studentized_tailprob(K, q, 'order', M)                 % P(Q >= q)
r = studentized_tailprob(K, q, 'order', M, 'lower', true)  % P(Q <= q)
r = studentized_quantile(K, p, 'order', M)                 % the q with P(Q <= q) = p
sd = studentized_saddlepoint(K, q)                         % sd.s, sd.t, sd.v, sd.q0
loc = studentized_locus(K, s, tguess)                      % loc.t, loc.v, loc.q
```

Unlike a ratio, $X - q\sqrt{Y}$ is not linear in $(X, Y)$, and the inversion formula is a double integral. Its saddlepoint $(\hat{s}, \hat{t}, \hat{v})$ solves

$$K_s(s,t) = qv, \qquad K_t(s,t) = v^2, \qquad qs + 2tv = 0,$$

with $\hat{v} > 0$; `studentized_saddlepoint` finds it by Newton's method with continuation in $q$ from $q_0 = E(X)/E(Y)^{1/2}$, where $\hat{s} = \hat{t} = 0$. The density is the amplitude expansion of the double integral about the saddlepoint, and the distribution function an expansion of Hauschildt–Robinson type in $s$ after the $t$-integral is expanded by Laplace's method (§ 10.5). `r.f(M)` keeps the terms of the density of orders $n^{-m}$, $m \le M - 1$, and `r.F(M)` those of the distribution function of orders $n^{-m/2}$, $m \le 2M - 1$, both with relative error $O(n^{-M})$; `r.info.Fp(j+1)`, $j = 0, \dots, 2M - 1$, holds the distribution function with the terms up to $n^{-j/2}$. `r.saddle` is the saddlepoint and `r.info.kappa` the cancellation indicator. The option `'saddle', sd` supplies a saddlepoint already computed. The expansions are not evaluated at $q = q_0$, where $\hat{t} = 0$.

The saddlepoints of all $q$ lie on the curve $sK_s + 2tK_t = 0$ through the origin, on which $v = K_t^{1/2}$ and $q = K_s/v$ (`studentized_locus`). `studentized_quantile` searches along this curve: each trial value of $\hat{s}$ gives its $q$ and its distribution function directly, so that no saddlepoint equation is solved for a trial $q$ (a bracket in $\hat{s}$ from the origin, then the Illinois iteration; `'tol'`, default $10^{-10}$, is the tolerance on the probability).

**Example: the noncentral $t$ distribution.** With $X \sim N(\mu, 1)$ and $Y \sim \chi^2_\nu/\nu$ independent and $n$ pairs, $\bar{X}/\sqrt{\bar{Y}} = T/\sqrt{n}$, $T$ noncentral $t$ with $n\nu$ degrees of freedom and noncentrality $\sqrt{n}\mu$:

```
mu = 1; nu = 3; n = 16;
K = sample_mean(CGF(@(t) mu*t(1) + t(1)^2/2 - nu/2*log(1 - 2*t(2)/nu), 2, ...
                    'domain', @(t) t(2) < nu/2), n);
studentized_tailprob(K, 6/sqrt(n), 'order', 3, 'lower', true).F'   % P(T <= 6)
%   0.9539837462   0.9539799842   0.9539798477    (exact 0.9539798478)
studentized_density(K, 6/sqrt(n), 'order', 3).f'/sqrt(n)           % density of T at 6
%   0.0757466061   0.0753761567   0.0753767005    (exact 0.0753767236)
studentized_quantile(K, 0.05, 'order', 3).q*sqrt(n)   % 2.32153317, the exact value
```

**Example: the Dickey–Fuller $t$-ratio.** `tests/dickey_fuller_cgf.m` gives the joint cumulant generating function of $(U, V)$ whose ratio $U/\sqrt{V}$ is the limiting Dickey–Fuller $t$-ratio without deterministic terms (model 1), with a constant (model 2) or with a constant and a trend (model 3), under the unit root ($c = 0$) and local alternatives ($c < 0$), from Nabeya and Tanaka (1990) and Nabeya and Sørensen (1994). For model 2 at the 5% critical value,

```
K = dickey_fuller_cgf(2);
r = studentized_tailprob(K, -2.862, 'order', 3, 'lower', true);  r.info.Fp'
%   0.0475  0.0520  0.0484  0.0449  0.0453  0.0496
studentized_quantile(K, 0.05, 'order', 3).q                      % -2.85930
```

(the exact quantile of the limit is $-2.861569$). It is a test fixture, not part of the package.

**Precision.** Where $|\hat{t}|$ is large the terms of high order of the density cancel by many orders of magnitude (for the Dickey–Fuller model 2 above the indicator is about $10^8$ at order three, $10^{13}$ at order four and $10^{17}$ at order five). When `r.info.kappa(end)` exceeds $10^6$, `studentized_density` repeats the computation in double-double arithmetic (§ 9.1), as the Julia package repeats it in `BigFloat`: the saddlepoint is refined, and the Taylor polynomial of $K$, the whitening and all sums carry about 32 digits; `r.info.refined` is then `'double-double'`. The relative error of the refined values is about $\kappa \times 10^{-32}$: for the Dickey–Fuller model 2 the refined density agrees with the Julia package in `BigFloat` to the last digit of double precision up to order four and to $10^{-15}$ at order five; `r.status` remains `'coefficients inaccurate'` when $\kappa\, 2^{-104} > 10^{-10}$ (order six for that model). The options are

```
r = studentized_density(K, q, 'order', M, 'refine', false)    % double precision only
r = studentized_density(K, q, 'order', M, 'precision', 'dd')  % double-double from the
    start
r = studentized_tailprob(K, q, 'order', M, 'precision', 'dd')
```

without refinement `r.status` is `'coefficients inaccurate'` when the indicator exceeds $10^{10}$, and `'refinement failed'` if $K$ cannot be evaluated on double-double series. The distribution function is much less affected; in double-double it costs a few seconds, mostly in the integrals $I_m$, which are computed in variable precision. A refined density of order six costs about ten seconds.

# 9 Derivatives, saddlepoints and the Taylor class

```
D = derivatives(K, t, N)       % all derivatives of orders <= N at t
[tau, info] = saddlepoint(K, x)
p = taylor(K, t, N, M)         % Taylor polynomial of K(t + M v) in v, degree N
```

`D.nu` holds the multi-indices $\nu$, one per row in graded order, and `D.value` the derivatives $K^{(\nu)}(t)$, so that `D.value(1)` is $K(t)$. The saddlepoint is computed by Newton's method with backtracking on the convex function $K(\tau) - \tau^{\mathsf{T}} x$ (`spa_saddle`).

The class `tps` represents arrays of polynomials in $k$ variables truncated at total degree $N$:

```
v = tps.vars(k, N)       % k x 1 array of the variables
p = exp(v(1)) * log(1 + v(2))
coef(p, [1 2])           % nu! times the coefficient of v^nu, the derivative at 0
double(p)                % the constant term
T = tps.tables(k, N);    % T.mi: the multi-indices in graded order
```

A function $f$ of a series $p = p_0 + q$, $q$ without constant term, is evaluated by composition, $f(p) = \sum_{j=0}^{N} f^{(j)}(p_0) q^j / j!$; where $f$ is not analytic at $p_0$ an error with identifier `'tps:domain'` is raised. The coefficients are stored in `p.C`, one column per element of the array.

### 9.1 Double-double arithmetic

The class `dd` holds a number as the unevaluated sum $x_{\mathrm{hi}} + x_{\mathrm{lo}}$ of two doubles with $|x_{\mathrm{lo}}| \le \frac{1}{2}\mathrm{ulp}(x_{\mathrm{hi}})$, about 32 significant digits, with the error-free transformations of Knuth and Dekker (two-sum, two-product by splitting). Addition, multiplication, division and the square root are computed in this arithmetic, sums are compensated and pairwise; the other elementary functions (`exp`, `log`, `erfc`, ...) are evaluated through `vpa` with 40 digits and rounded back.

```
x = dd(sym(1)/3)            % from a symbolic number, a double, or (hi, lo)
y = sqrt(x)*x + 1;  vpa(y, 32)   % double(y) rounds to double
```

The class `tpsdd` is the Taylor class in double-double: the same interface as `tps` (`tpsdd.vars`, `tpsdd.const`, `coef`, arithmetic, `exp`, `log`, `sqrt`, `atan`, `sinhc`, ...), with the coefficients in `p.H` and `p.L`; `coef` returns a `dd`. The derivatives of a function at the expansion point, needed for the composition, are computed in `vpa`. `taylordd(K, t, N, M)` is the Taylor polynomial of $K$ in double-double; it evaluates the function of `K` on `tpsdd` series, so it applies to the cumulant generating functions written with these operations (the library, the transformations, and `tests/dickey_fuller_cgf.m`), but not to a closed-form `'Taylor'` function, which works in double precision. A product of two bivariate series of degree 30 takes about ten milliseconds, several orders of magnitude less than the same product in `vpa`. The Symbolic Math Toolbox is needed for the constants and the elementary functions.

## 10 The expansions

This section states what the package computes. The paper gives the derivations, the conditions and the error bounds; here $n$ is absorbed in $K$.

### 10.1 Whitened coefficients

Let $\hat\tau$ be the saddlepoint, $K'(\hat\tau) = x$, and $K''(\hat\tau) = LL^{\mathsf T}$ the Cholesky factorization. All terms are computed for the whitened vector $Y = L^{-1}X$, whose cumulant generating function $K_Y(u) = K(L^{-\mathsf T}u)$ has its saddlepoint at $\hat u = L^{\mathsf T}\hat\tau$ and Hessian $I$ there. Let $\theta_\nu = K_Y^{(\nu)}(\hat u)/\nu!$, $|\nu| \ge 3$, be its Taylor coefficients and define $c_{r,\nu}$ by

$$c_{1,\nu} = \theta_\nu, \qquad c_{r,\nu} = \frac{1}{r}\sum_{k=3}^{|\nu|-3(r-1)} \sum_{|\lambda|=k,\ \lambda\le\nu} \theta_\lambda\, c_{r-1,\nu-\lambda}, \qquad r \ge 2,$$

so that $c_{r,\nu}$ is the coefficient of $u^\nu$ in $\Theta(u)^r/r!$, where

$$\Theta(u) = \sum_{k=3}^{\infty} \sum_{|\nu|=k} \theta_\nu u^\nu.$$

The multivariate Hermite polynomials at zero factor into univariate ones,

$$\mathrm{He}_\nu(0) = \prod_{j=1}^{d} \mathrm{He}_{\nu_j}(0), \qquad \mathrm{He}_{2i}(0) = (-1)^i \prod_{j=1}^{i}(2j-1), \qquad \mathrm{He}_{2i+1}(0) = 0.$$

The package obtains the $\theta_\nu$ by differentiating the composed function $K(\hat\tau + L^{-\mathsf T}u)$ directly. In the original coordinates the terms cancel much more when the components of $X$ are strongly correlated under the tilted distribution, and the substitution of $L^{-\mathsf T}$ into the Taylor polynomial of $K$ can lose many digits at high degree.

## 10.2 Densities

$$f(x) \approx \frac{\exp\{K(\hat\tau) - \hat\tau^{\mathsf T} x\}}{(2\pi)^{d/2}|K''(\hat\tau)|^{1/2}} \sum_{m=0}^{M-1} a_m, \qquad a_0 = 1, \qquad a_m = \sum_{r=1}^{2m} \sum_{|\nu|=2m+2r} c_{r,\nu} \mathrm{He}_\nu(0).$$

The coefficient $a_m$ involves the $\theta_\nu$ with $|\nu| \le 2m+2$.

## 10.3 Orthant probabilities

For $\hat\tau \ge 0$,

$$\mathrm{P}(X \ge x) \approx \exp\{K(\hat\tau) - \hat\tau^{\mathsf T} x + |\hat u|^2/2\} \sum_{m=0}^{2M-1} T_m, \qquad T_0 = \tilde I_0, \qquad T_m = \sum_{r=1}^{m} \sum_{|\nu|=m+2r} c_{r,\nu} \tilde I_\nu,$$

with

$$\tilde I_\nu = \int_{\{v \in \mathbb{R}^d:\ Lv \ge 0\}} \prod_{j=1}^{d} \mathrm{He}_{\nu_j}(v_j)\, \phi_d(v + \hat u)\, dv,$$

$\phi_d$ the standard normal density in $d$ dimensions. The term $T_m$ is of order $n^{-m/2}$. When $\hat\tau$ has negative components, let $\mathcal{N} = \{j : \hat\tau_j < 0\}$. Then

$$\mathrm{P}(X \ge x) = \sum_{T \subseteq \mathcal{N}} (-1)^{|T|}\, \mathrm{P}\big(\tilde X_{\mathcal{N}^c \cup T} \ge \tilde x_{\mathcal{N}^c \cup T}\big),$$

where $\tilde X$ and $\tilde x$ are $X$ and $x$ with the components in $T$ negated and the subscript selects components; the term with $\mathcal{N}^c \cup T$ empty is one. Each term has a saddlepoint in its closed positive orthant, or is decomposed again.

## 10.4 Ratios

Let $\mathcal{K}(s,t)$, $s,t \in \mathbb{R}^d$, be the joint cumulant generating function of $(X,Y)$, $W_j = X_j - r_j Y_j$, and $r \circ s = (r_1 s_1, \ldots, r_d s_d)$. The cumulant generating function of $W$ is $K_W(s) = \mathcal{K}(s, -r \circ s)$, and $\mathrm{P}(R \le r) = \mathrm{P}(W \le 0)$ is the orthant probability of $-W$ at zero. For the density, let

$$\mathcal{K}_B = \frac{\partial^{|B|}\mathcal{K}}{\prod_{j\in B} \partial t_j}, \qquad B \subseteq \{1, \ldots, d\},$$

and let $\mathcal{P}_{d,j}$ be the set of partitions of $\{1,\ldots,d\}$ into $j$ blocks and

$$g_k(s) = \sum_{\pi \in \mathcal{P}_{d,d-k}} \prod_{B \in \pi} \mathcal{K}_B(s, -r \circ s), \qquad k = 0, \ldots, d-1.$$

With $\hat s$ the saddlepoint of $K_W$, $K_W'(\hat s) = 0$, and $K_W''(\hat s) = LL^{\mathsf T}$,

$$f_R(r) \approx \frac{\exp\{K_W(\hat s)\}}{(2\pi)^{d/2}|K_W''(\hat s)|^{1/2}} \sum_{m=0}^{M-1} \tilde a_m, \qquad \tilde a_m = \sum_{k=0}^{\min(m,d-1)} b_{m-k}(g_k),$$

$$b_m(h) = \sum_{r=0}^{2m} \sum_{j=3r}^{2m+2r} \sum_{|\nu|=j} \sum_{|\mu|=2m+2r-j} c_{r,\nu}\, \gamma_\mu\, \mathrm{He}_{\nu+\mu}(0), \qquad c_{0,0} = 1,$$

where $\theta_\nu$, and hence $c_{r,\nu}$, are the whitened coefficients of $K_W$ at $\hat s$ and $\gamma_\mu$ the Taylor coefficients of $h(\hat s + L^{-\mathsf T} u)$ in $u$. The leading coefficient is $\tilde a_0 = g_0(\hat s) = \prod_{j=1}^{d} \mathcal{K}_{\{j\}}(\hat s, -r \circ \hat s)$. All the coefficients come from one Taylor polynomial of

$$\mathcal{K}\big(\hat s + L^{-\mathsf T} u,\ -r \circ (\hat s + L^{-\mathsf T} u) + \tau\big)$$

in $(u,\tau)$: the coefficient of $\prod_{j\in B} \tau_j$ is the Taylor polynomial of $\mathcal{K}_B$ in $u$. The sums $b_m$ are evaluated as He-weighted sums of the coefficients of the products $\Theta(u)^r\, h(u)/r!$.

### 10.5 Studentized means

For $Q = X/\sqrt{Y}$ the inversion formulas of Bao, Broda and Kan are double integrals,

$$f_Q(q) = \frac{1}{(2\pi i)^2}\iint e^{K(s,t)}\,2J_2(qs,t)\,dt\,ds, \qquad \mathrm{P}(Q>q) = \frac{1}{(2\pi i)^2}\iint e^{K(s,t)}\,2J_1(qs,t)\,dt\,\frac{ds}{s},$$

with the kernels $J_k(b,t) = \int_0^\infty v^k e^{-bv-tv^2}\,dv$, whose exponent is $q^2s^2/(4t)$ plus Gaussian terms. With $\psi(s,t) = K(s,t) + q^2s^2/(4t)$ the density is

$$f_Q(q) \approx \frac{\hat v^2 e^{\psi(\hat s,\hat t)}}{\sqrt{\pi}\,\{\hat t\,|\psi''(\hat s,\hat t)|\}^{1/2}}\Big\{1 + \sum_{m=1}^{M-1}\big(b_m(h_0) + b_{m-1}(h_1)\big)\Big\},$$

the amplitude expansion of § 10.4 in two variables with the amplitudes $h_0 = (s/\hat s)^2(t/\hat t)^{-5/2}$ and $h_1 = (t/\hat t)^{-3/2}/(2\hat t\hat v^2)$; the Hessian $\psi''$ is indefinite where $\hat t < 0$, and the Hermite moments are evaluated in the coordinates of a complex factorization $\psi'' = LL^{\mathsf T}$ from the eigen-decomposition $\psi'' = V\Lambda V^{\mathsf T}$. The double-double computation uses the real whitening $W = V|\Lambda|^{-1/2}$, $W^{\mathsf T}\psi''W = \mathrm{diag}(\sigma)$ with $\sigma_i = \pm 1$, and the Hermite moments of the indefinite form, $\prod_i (-1)^{n_i/2}\sigma_i^{n_i/2}(n_i-1)!!$, which gives terms of the same magnitudes. For the distribution function the $t$-integral is expanded by Laplace's method, which leaves a univariate integral of Hauschildt–Robinson type in $s$ with the exponent $\Psi(s) = \psi(s,t(s))$, $\psi_t(s,t(s)) = 0$, and amplitudes $H_k(s)$; the Taylor coefficients of $\Psi$ and of the $H_k$ at $\hat s$ are obtained from the Taylor polynomial of $K$ at $(\hat s,\hat t)$ by Newton's method and composition on power series, exact up to rounding.

## 11 Accuracy and precision

The computations are in double precision, except the double-double refinement of the density of a Studentized mean (§ 9.1). Rounding errors are about $10^{-14}$ relative in the tests, far below the asymptotic errors of the expansions at the orders used in practice ($10^{-3}$ to $10^{-9}$), so double precision suffices for nearly all purposes. It matters in three situations:

- *Cancellation in three or more dimensions.* At high orders the terms can cancel by many orders of magnitude. The engine reports this (`kappa` in `info`) and, when the cancellation exceeds $10^6$, tries to repeat the computation in variable precision. The Taylor class `tps` works in double precision, and its double-double counterpart is used only for Studentized means, so in this package that refinement is not possible: the double-precision values are kept and `status` reports `'refinement failed'`. The Julia package repeats the computation in `BigFloat`.
- *Studentized means.* The density of a Studentized mean can cancel strongly where the saddlepoint has large $|\hat t|$; it is then recomputed in double-double arithmetic (§§ 8 and 9.1), and `status` reports what remains.
- *Probabilities below about* $10^{-308}$. In two dimensions use `'logscale'` (§ 5).
- *Tables of very small relative errors*, at large $n$ and high order, need exact values and expansions beyond 15 digits. The engine programs accept derivatives in variable precision (`vpa`, Symbolic Math Toolbox) when they are supplied in closed form (§ 12).

## 12 The engine

The folder `engine` contains the programs of the paper. They take the derivatives of $K$ from a function `Kfun(nu, t)`, the derivative of order `nu` ($1 \times d$) at `t` ($1 \times d$), `NaN` outside the domain, and can be used directly when closed-form derivatives are available:

| | |
|---|---|
| `spa_saddle` | saddlepoint by Newton's method with backtracking |
| `spa_uni` | Daniels and Hauschildt–Robinson expansions from $K^{(k)}(\hat{\tau})$ |
| `spa_ratio_pdf` | Rice's expansion of the density of a ratio |
| `spa_bi`, `spa_orthant2` | bivariate density and orthant probability |
| `spa_tri`, `spa_mv`, `spa_orthant` | three and more dimensions |
| `qf_Kmulti` | derivatives of the cgf of quadratic forms |
| `ncx2_spa`, `ncf_spa`, `dncf_spa` | noncentral $\chi^2$, $F$ and doubly noncentral $F$ |
| `spa_uni_hp`, `spa_bi_hp` | variable precision |

The class `CGF` builds `Kfun` and the whitened derivatives (`opts.Kwhite`) from the Taylor polynomials. See the help of each program.

## 13 Validation

`tests/test_spx.m` runs the following checks and returns `true` if all pass (`test_spx(true)` prints every result):

- derivatives of the Taylor class against closed forms and finite differences, and the derivatives of a pair of quadratic forms by automatic differentiation through a generic Cholesky factorization against the trace recursion;
- the 409 reference values shared by the Julia and Python packages (`tests/julia_reference.csv`, generated by `test/reference_values.jl` of the Julia package; `tests/julia_reference.jl` copies them): derivatives, saddlepoints, orthant probabilities with lower and mixed corners and saddlepoints with negative components, and densities in one to four dimensions (sums of exponentials including the examples of Wang and of Kolassa and Li, gamma, noncentral chi-squared, uniform, common-component, quadratic-form, Wishart and normal distributions), single ratios (the noncentral $F$) and several ratios. They agree to about $10^{-13}$ relative, at most $10^{-11}$; the four-dimensional orthant probabilities, computed by quasi-Monte Carlo, agree within three standard errors;
- the expansion for several ratios at $d = 1$ against Rice's expansion ($10^{-14}$), and for two and three ratios against the inverted Dirichlet density and distribution function and the product of two independent beta-prime ratios, including the rate $O(n^{-4})$ of the density of order four;
- the tests of the Studentized-mean functions of the Julia and Python packages: the noncentral $t$ distribution against exact values (distribution function to $2 \times 10^{-9}$ at order three, density to $10^{-6}$, the quantile and the saddlepoint locus), and the Dickey–Fuller $t$-ratio against the reference values of Bao, Broda and Kan (the distribution function with every half-order to $10^{-8}$, the density to $2 \times 10^{-6}$, a saddlepoint and a probability under a local alternative, and the 5% quantile of model 2). The values agree with those of the Python package `saddlepoint-expansions` 0.2.2 to about $10^{-13}$ for the noncentral $t$ and to $10^{-11}$–$10^{-9}$ for the Dickey–Fuller model, whose derivatives of high order carry rounding errors;
- double-double arithmetic: `dd` against `vpa` (about $10^{-32}$), the derivatives of a `tpsdd` series against symbolic differentiation (about $10^{-32}$), and the Studentized expansions in double-double against the Julia package in `BigFloat` with 256 bits (`tests/studentized_reference.csv`, generated by `tests/julia_studentized_reference.jl`): the density of orders one to six and the distribution function with every half-order up to order four, for the noncentral $t$ at

$q = 0.5$ and the Dickey–Fuller model under a local alternative, agree to the last digit of double precision, and the refined density of the Dickey–Fuller model 2 to $10^{-15}$ at order five. Over all the reference values (the noncentral $t$ at $q = 0.5, 1.5, 2.5$ and the three Dickey–Fuller models), double-double agrees to the last digit except the density of model 2 at order five ($10^{-15}$) and order six ($4 \times 10^{-10}$, $\kappa \approx 10^{23}$); for that model double precision has errors of $2 \times 10^{-9}$ at order three and gives no correct digit at order six.

In addition, for all 24 points of the table of sample autocorrelations of the paper, the density and distribution function of two ratios computed by the package agree with those of the dedicated program of the paper (`acf_spa`) to $1.5 \times 10^{-12}$ at every order from one to five.

## 14 Function reference

| call | result |
|---|---|
| `K = CGF(fun, d, 'domain', pred, 'Taylor', tfun)` | a cumulant generating function |
| `r = tailprob(K, x, 'order', M, 'lower', tf, 'signs', s, ...)` | orthant probabilities; `r.F`, `r.tau`, `r.status`, `r.info` |
| `r = density(K, x, 'order', M, ...)` | densities; `r.f`, `r.tau`, `r.status`, `r.info` |
| `r = ratio_density(K, r0, 'order', M)` | densities of $d =$ `numel(r0)` ratios; `r.f`, `r.shat`, `r.info` (for $d \geq 2$: the terms `a`, the log prefactor `logC`) |
| `r = ratio_tailprob(K, r0, 'order', M, 'lower', tf, 'signs', s, ...)` | distribution functions of ratios; `r.F`, `r.shat`, `r.status`, `r.info` |
| `r = studentized_density(K, q, 'order', M, 'saddle', sd, 'refine', tf, 'precision', p)` | density of $X/\sqrt{Y}$; `r.f`, `r.saddle`, `r.status`, `r.info` (`'refine'`, default `true`: double-double when $\kappa > 10^6$; `p`: `'double'` or `'dd'`) |
| `r = studentized_tailprob(K, q, 'order', M, 'lower', tf, 'saddle', sd, 'precision', p)` | distribution function of $X/\sqrt{Y}$; `r.F`, `r.saddle`, `r.status`, `r.info.Fp`, `r.info.kappa` |
| `r = studentized_quantile(K, p, 'order', M, 'tol', tol)` | quantile of the expansion; `r.q`, `r.saddle` |
| `sd = studentized_saddlepoint(K, q)`, `studentized_locus(K, s, tguess)` | the saddlepoint and the saddlepoint locus |
| `[tau, info] = saddlepoint(K, x)` | the saddlepoint |
| `D = derivatives(K, t, N)` | `D.nu`, `D.value` |
| `p = taylor(K, t, N, M)` | Taylor polynomial of $K(t + Mv)$ |
| `v = eval(K, t)` | $K(t)$ |
| `reflect`, `marginal`, `linear_combination`, `sample_mean`, `shift`, `+` | transformations (§ 4) |
| `tps.vars(k, N)`, `tps.const(c, k, N)`, `coef(p, nu)`, `double(p)`, `tps.tables(k, N)` | the Taylor class |
| `x = dd(a)`, `dd(hi, lo)`, `vpa(x, n)`, `double(x)` | double-double numbers (§ 9.1) |

| call | result |
|---|---|
| `tpsdd.vars(k, N)`, `tpsdd.const(c, k, N)`, `tpsdd.fromtps(p)`, `coef(p, nu)`, `ddconst(p)` | the Taylor class in double-double |
| `p = taylordd(K, t, N, M)` | Taylor polynomial of $K(t + Mv)$ in double-double |
| `sinhc(x)`, `sincu(x)` | $\sinh(x)/x$ and $\sin(x)/x$ |
| `spx_setup` | adds the package to the path |

# References


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